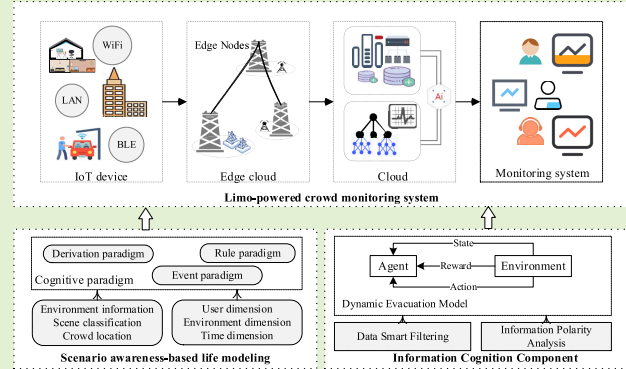


The Limo-Powered Crowd Monitoring System: Deep Life Modeling for Dynamic Crowd With Edge-Based Information Cognition

Rui Wang¹, Qiao Yu¹, Bander Alzahrani², Ahmed Barnawi¹,
Ahmed Alhindi¹, *Senior Member, IEEE*, and Miaomiao Zhao¹

Abstract—To ensure the safety of people in massively crowded activities, crowd monitoring systems are very important. Presently, many researches focus on studying video for crowd monitoring, and realize the recognition of behavior patterns. However, most systems only implement a real-time display of the monitored scene to comprehensively model a place. Because the environment at crowd gatherings is complex and dynamic, an intelligent monitoring system is necessary. This paper proposes a limo-powered crowd monitoring system for modeling dynamic crowds. On one hand, limo-system realizes life modeling at the individual level, offering personalized cognition of each person. On the other hand, it uses data collected by multi-sensor devices and intelligent algorithms deployed on the edge cloud to dynamically perceive the environment. At the same time, considering the redundancy of monitoring data, the value of information can be recognized via information cognition components in the edge cloud. Finally, combining the results of environment awareness and life modeling, simulation experiments of crowd scenes are executed for multi-scenarios. Results show that the intelligent system can be effectively applied to crowd monitoring in comprehensive environments, with important significance for crowd perception and modeling.

Index Terms—Crowd monitoring, scene awareness, life modeling, information cognition.



I. INTRODUCTION

POPULATION growth and continuously increasing social activities undoubtedly bring potential risks to public safety. It is very important to monitor the places where massive crowds gather. With the application of 5G and increasingly digital cities, monitoring devices such as CCTV or camera deployed on unmanned aerial vehicles (UAV) are spread

all over, which generates massive monitoring data full of information [1], [2]. Once the video stream is processed and analyzed by intelligent algorithms, it helps manage incidents to ensure public safety [3].

With the exponential growth of video data, an automated method must be used to monitor and simulate crowds, and an intelligent video analysis is particularly critical. In recent years, with the development of deep learning, many researchers have focused on extracting high-dimensional features from video stream using deep neural network processing and then realizing crowd modeling [4], [5]. Haroon *et al.* solved the issues of counting, density-map estimation and crowd positioning for a given high-density scene image [6]. Rei *et al.* used Online Sequential Extreme Learning Machine (OS-ELM) and an auto-encoder to carry out adaptive abnormal behavior detection [7]. Wenxi *et al.* proposed a method that used multiple exemplar agent-based motion models to extract motion features from crowd trajectories. It combined overall features to execute personal features to describe crowd motion. This method has shown better results in real-world crowd scene classification [8]. To solve the problem of head detection under unconstrained posture and direction conditions, Wei *et al.* constructed an adaptive relational network to

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Rui Wang, Qiao Yu, and Miaomiao Zhao are with the School of Computer Science and Technology, Huazhong University of Science and Technology, Wuhan 430074, China (e-mail: ruiwang2020@hust.edu.cn; qiaoyu_epic@hust.edu.cn; miaomiao0220@hust.edu.cn).

Bander Alzahrani and Ahmed Barnawi are with the Faculty of Computing and Information Technology, King Abdulaziz University, Jeddah, Saudi Arabia (e-mail: baalzahrani@kau.edu.sa; ambarnawi@kau.edu.sa).

Ahmed Alhindi is with the Department of Computer Science, Umm Al-Qura University, Mecca, Saudi Arabia (e-mail: ahhindi@uqu.edu.sa). Digital Object Identifier 10.1109/JSEN.2021.3080917

capture contextual information. By fully adaptively extracting global and local information, the model can be integrated into an end-to-end head detector [9].

The mentioned relevant studies on crowd analysis mainly focus on extracting local or global features of the crowd, analyzing the sequence of video stream or static images, and studying the crowd's characteristics in the scene [10]–[12]. In fact, most existing crowd analysis scenarios are designed for specific application scenarios, far from considering crowd modeling in the complex activities. Therefore, the scope of crowd modeling's applications is narrow, which make it difficult to apply it to real-time crowd monitoring systems [13], [14]. In addition, most monitoring systems can only display frames of the scene but do not perform in-depth modeling or provide valuable information for observers. At the same time, it is labor-intensive to observe the video stream of complex scenes. Therefore, if only single-layer data are used to train the deep learning model used for intelligent crowd scene modeling, this monitoring system cannot meet the service needs of real scenes. At present, the difficulties of intelligent monitoring system are as follows:

- Single scene and data sources of monitoring applications: although many researchers are involved in crowd modeling research, they often focus on single-scene analyses, such as abnormal behavior identification, crowd counting, crowd location, and crowd tracking. In a single scenario, it is difficult to combine multi-task work for in-depth crowd modeling. At the same time, single data sources also limit the expansion of multi-scenario modeling, making it difficult to realize multi-scenario joint recognition and simulation.
- Low feedback interaction for monitoring system results: when an observer uses a system to monitor a specific scene, many video streams are often time-consuming when finding valuable and specific information. This time-consuming and labor-intensive observation strongly affects the level of crisis warning and intervention. The cost of human resources results in a dissatisfactory experience, and thus cannot attract more users to empower big data. Besides, the intelligence level is developed in a single direction, and there is a lack of understanding of where the artificial intelligence (AI) technology should be expanded from a single field to multiple fields.
- Weak personalization degree of crowds: when modeling crowds, it is important to accurately capture the characteristics of the crowd, which cannot be ignored. In large-scale crowds, often only part of the people disturb the normal and cause serious losses of life and property. If individuals' characteristics are ignored in the modeling, it will cause huge hidden dangers to the safety of the crowd. But in the crowds, especially in large-scale crowd, realizing personalized individual modeling is very complex.

The main limitations of the existing crowd monitoring systems (called “low-intelligence systems”) are listed. However, the high-intelligence system in the monitoring field represents a highly intelligent platform with information cognition capabilities and deep individual simulation modeling [15], [16].

The basic limitation that hinders developing low-intelligent system into high-intelligent platforms is the fact that surveillance video data have a single source, which lacks multidimensional data and complex consideration of real application scenarios. For example, in large-scale crowd activities, Almeida *et al.* [17] only described the speed and direction of crowd movement, which cannot capture individuals' behavior. At the same time, when a small-scale crowd abnormality occurs, the crowd monitoring system conduct make any early warning or diversion tasks. On the one hand, when the model based on a single scenario is migrated to multiple scenarios, it is difficult to provide effective services in the new environment, and it comes with huge migration costs. On the other hand, most monitoring systems cannot respond to the abnormal behaviors of the population in real time. Therefore, many existing monitoring systems cannot fully consider every complex situation in the actual service scenario.

In fact, the crowd comprises many independent individuals or sub crowds. If accurate cognition for the individual can be achieved, everything will become clearer in the analysis and modeling of the crowd. Therefore, we must combine multi-source data to achieve information fusion and then perform information cognition to complete the population modeling. In recent years, intelligent fabric technology has made great breakthroughs [18]. The appearance and the tactile impression of smart textiles made of intelligent fabric are almost the same as the traditional fabric. This kind of fabric is an electronics system acts as a data-processing module. A data network powered by intelligent fabric can access many aspects of human life and collect multidimensional data. Intelligent fabric is a kind of signal collector made of flexible electronic materials. By making it into clothes, the body characteristics can be analyzed via signals collected from the fabric throughout the body for behavior modeling. At the same time, other sensors deployed on the intelligent fabric can also be used to collect human physiological index signals.

Therefore, based on the intelligent fabric and bionic technology, we propose a new platform called the limo-powered crowd monitoring system, which represents a highly intelligent system. The term “limo-powered” means “life modeling”. The data collected from the real world are used to model human life thoroughly, and these data can support the limo-robot's implementation of a high-resolution humanoid. However, the computing and storage resources of any equipment are limited. In such a multidimensional and large-scale data collection and processing system, we must consider information filtering. Otherwise, there will be a great challenge related to the communication bandwidth and resource storage for the whole system. In this work, we consider the three following aspects in the ecosystem design.

- **Data network-based intelligent fabric:** The limo system collects the data of human beings from the physical world and builds a deep-level life model on the remote cloud. The data collected by the intelligent fabric overcome barriers of the traditional data collection method to offer an unobtrusive data collection process, which has high significance. Meanwhile, other data are collected from multi-sensors to join data network.

- **Deep life modeling database:** The remote cloud has rich storage and computing resources that can be used to store the data, including personalized movements, emotions, and behaviors of people in a different environment to deeply understand human life. The database can support the limo-robot high-resolution humanoid to become a human incarnation.
- **Intelligent information cognitive components:** These components can cognize the data that flow through the system at front. Near the device's end, deficient data are filtered in advance, and the communication load of the backbone network is reduced. On the remote cloud, the data training model with a positive significance to society is chosen using the information classification methods. While interacting with platforms and people, attention is paid to the perception and understanding of emotions to obtain the effect of empathy with people [19], [20].

Overall, the proposed limo-system represents a completely new monitoring form that has not been reported yet, to the best of our knowledge. By using intelligent fabric technology, an intelligent monitoring system is designed. Concretely, the smart textile collects the physical data to build a precise model of human life. The main contributions of this paper are as follows:

- **An ecosystem of the crowd-monitoring limo system is introduced.** This system mainly contains multi-sensors and an intelligent-fabric data-based network, edge nodes of information cognition, intelligent remote cloud database, and a crowd-monitoring system capable of life modeling humans.
- **The scenario awareness-based life modeling is presented.** The image and sensor data deployed in the environment are integrated to recognize the scenario accurately. The complex characteristics of social life are also considered, and a multidimensional life database is established.
- **Information components for massive monitoring data are designed.** Based on cognitive information theory, the data filtering, information polarity analysis, and dynamic evacuation model are formulated to deal with the real-time data stream in complex application scenarios.

The rest of this paper is organized as follows: Section II establishes the architecture of the limo system. Section III introduces the scenario awareness model and the design of the limo database. Section IV analyzes the design issues combined with cognitive information theory related to the data flow used in the ecosystem. In Section V, experiments are performed, including crowd modeling and simulations to evaluate the feasibility of the proposed scheme. Finally, we conclude the paper in Section VI.

II. SYSTEM ARCHITECTURE

This paper presents the limo-powered crowd monitoring system by deeply modeling the life of human beings and by mapping the life and behavior mode of each person to a virtual agent. The ultimate goal is to reconstruct the whole life of

each human and of the whole crowd. The architecture of the limo system appears in Fig. 1, including the data collection module, edge nodes, intelligent remote cloud, and crowd monitoring terminal. Furthermore, accurate construction, intelligent application, and limited resource domain support of a limo system are considered in the framework design. Concretely, an intelligent fabric-based data network, which represents the data collection layer of the system, can collect life data continuously, including the multidimensional data with time, space, the society attributes, and more. The edge nodes of the information cognition transmit data between the near-end data platform and the remote cloud platform [21]. As a data transfer station, the base stations deployed on edges provide the information cognition service [22] and lightweight heterogeneous or redundant data. Powerful computing and storage resources are deployed on the remote cloud to model human life deeply. Each agent in the limo system embodies a human with a deep-level life model to improve accuracy and personalization in crowd monitoring.

A. Intelligent Fabric-Based Data Network

Different from the traditional method where intelligent terminals collect data, the data-collection layer constructed via intelligent fabric has the spatial-temporal correlation and multidimensional characteristics. Intelligent fabric can be divided into environment fabric and wearable fabric.

First, intelligent fabric is spread across the whole living space of users to continuously collect data. Intelligent fibers are used in carpets, seats, table cloths, curtains, and accessories in public places like libraries, galleries, theaters, entertainment places, restaurants, cafes, and more. Multiple flexible micro sensors (e.g., the sitting posture sensor and voice collection device) are embedded in intelligent fibers to collect data including the user's behavior, sitting posture, and entertainment preferences. Based on the technology of intelligent fabric, the data collection process is not limited by the movability of human life. The Global Positioning System (GPS) sensors in textiles can record users' movement and collect real-time and spatial activity data along with quantity increases over time.

Because human life is very complex, physiology and emotion should also be considered. Moreover, a human cannot be separated from society. Thus, in addition to the environment fabric, we introduce intelligent wearable fabric in the proposed system. The textiles (intelligent clothes, intelligent hats, and intelligent bags) worn on the human body are close-fitting fabrics that can record the physiology, psychology, and joint data of users. When the user has a conversation, a micro camera and microphone record social information related to the user via video and audio data. With intelligent fabric technology, the physiology, psychology, daily behavior, and body movement of a user are collected in the data collection layer, which is the basis for building human life models and embodiment of crowds.

B. Edge Nodes of Information Cognition

Edge nodes of information cognition comprise the process of preprocessing data between the terminal data process

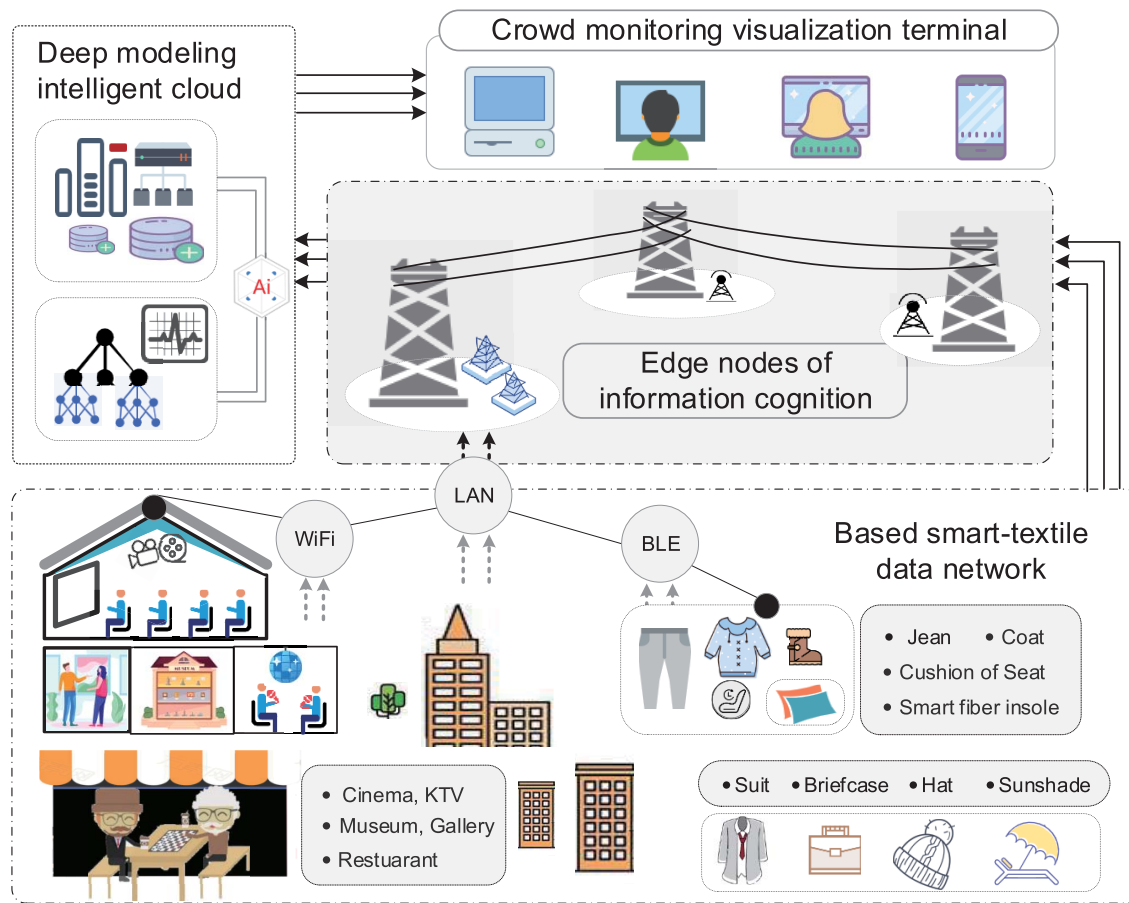


Fig. 1. The architecture of the limo-system.

and the remote intelligent cloud via a mobile cloudlet-assisted service mode named Opportunistic task scheduling over co-located clouds (OSCC) [23]. Considering the large quantity of terminal data and the limited communication and storage resources, the data near terminal position shall be filtered. Therefore, data processing will be performed at the edge base station. These data are transmitted from terminal devices to the edge network through a wireless network, which is responsible for transferring data. The main task of edge nodes is to receive, make the data compress, and package it to the cloud. As for the static data (such as stable physiological data and body posture during sleep), the healthy state of users can be analyzed at edge nodes without uploading to the cloud. Meanwhile, the process can also filter low-value data, and only the analysis result is uploaded to the cloud terminal to store. When data change, differences will be uploaded to a cloud terminal for further analysis. Some dynamic data, such as voice data, must be preprocessed. In order to avoid the network congestion caused by the real-time upload network, these data are packed and sent to remote clouds at regular intervals. The edge network receives a large quantity of user data from the terminal, which are preprocessed at edge nodes and then transferred to remote intelligent clouds.

C. Human Life Model on Remote Cloud

Considering that the life model of human individuals and crowds represents a multidimensional and complex model, the intelligent cloud is used to build the human life model. Due to its powerful computing and storage ability, the remote cloud can deeply analyze the data from the edge network and build the life model of each individual and of the whole crowd. The intelligent cloud is deployed with various AI algorithms, including the emotion recognition algorithms that explore the spiritual world of a user, physiological health-detection algorithms that evaluate the user-health level, and scene-awareness algorithms for scene recognition and crowd location that mine crowd behavior patten to build a user life model database from data with various dimensions. In the whole limo-system process, there are two stages of the intelligent cloud; the first stage is building the human life model. With the increase in data in the cloud, the human life model will upgrade gradually, and parameters of model will be optimized continuously. Consequently, the new information becomes more similar to what has been stored in the life model database until the model converges completely. The next stage is model upgrading. Namely, people have different habits and characteristics at different times. In different life stages, the actions, reactions, and social behavior of a person differ,

and sometimes that difference can be significant. Therefore, the life model developed on the cloud should learn and upgrade continuously by recording the user behavior and activity tracks to imitate users accurately in different life stages. However, the process of upgrading this model is slow.

D. The Crowd-Monitoring Visualization Terminal

Through data collection by the sensors, edge processing by the edge cloud, and AI-based deep modelling and cognition in the remote cloud, the system builds the life model of human individuals and of the crowd via a large amount of information (e.g., crowd position, individual behavior mode). To achieve crowd management, evacuation rehearsal, and other goals, the system needs to ultimately visualize such information in a user-friendly way via a terminal. The crowd-monitoring visualization system is a computer software system that displays the crowd and the environment in various forms (e.g., 3D animation, charts, and text). Take the gallery scene as an example. The smart carpet made of intelligent fibers can obtain the real-time position of each person. With the intelligent wearable fabric can record the postures and movement (e.g., walking, standing, head low, head up) of each person. Virtual agents at such positions and with such behaviors can be displayed as 3D animation in the visualization terminal. Besides, data obtained by sensors on the intelligent fabrics, such as physiological signal sensors (e.g., EEG sensors) and audio and video sensors can be used in AI cognition to mine information about people's health and emotion status [24], [25]. When the user hovers on certain virtual agent, their current emotional type and rating will be displayed. Virtual agents with physiological signals indicating risks (e.g., very low blood pressure or heart rate) or extreme anger will be highlighted for the managers, so that the risks can be controlled beforehand.

III. SCENE AWARENESS AND MULTIDIMENSIONAL LIFE MODELING

Because various scenes appear in crowd monitoring tasks, this section constructs a scene-perception model and a crowd-location model based on computer vision. Meanwhile, life model database containing several dimensions in each scenario for a user is established. Specifically, user behavior can be depicted in the time dimension and be recorded in the space dimension. This section introduces the scenario classification model and describes the design solution and the related components of the life-model database, which is based on the service scenarios.

A. Scene-Awareness Model

In current mainstream solutions, feature extraction of images and deep learning methods are used for single-modal scene classification. These methods require high-quality images, which is more applicable to coarse-grained classification and cannot classify scenes in a fine-grained manner. However, in the intelligent fabric-based data collection network, micro sensors upload user data to the remote cloud through edge nodes at defined intervals. Among these sensors, a camera in smart clothing can capture scene pictures

from a user perspective, and a GPS device can acquire the user's location information. Therefore, with intelligent-fabric technology, the scenario recognition model is developed in combination with environment data and images to recognize scenes with high precision.

The crowd location can be detected via environmental data. The environment data comprises the geographical position data obtained by the GPS sensors, namely the longitude and latitude coordinates. Then, this location is used as the location center, with r as the radius, to search for stores in the map database. According to the method of storage and the location attributes, we can obtain the probability vector $\vec{gps}$ of a scene where the user may be located. In addition, to obtain the specific location of the individual in the image, we employed the Recurrent Attentive Zooming (RAZ_Loc) algorithm to realize the positioning of the crowd, which is used to capture the fine-grained location information of the individual in the crowd [26], [27]. RAZ_Loc algorithm is a recurrent attentive zooming network that detects the blurred image region recurrently and enhances it to high resolution for re-detection. It can count crowds and locate the individual in the crowd, so it is very suitable for our application requirements.

The scene can be recognized based on images from CCTV video. The frame difference method can filter the similar frames and resize all the images to the same size of 224×224 pixels. Subsequently, the images are enhanced to improve their quality. We used the ResNet50 [28] network model as an image scene classification model to predict the scenario vector $\vec{image}$. Considering that the image quantity of each time window is not large, an open-source dataset such as Places365 can be used to pretrain the ResNet model and initialize its parameters.

Based on location or images, our model of scene awareness yields a different effect of recognition results in multiple scenarios. In order to make the scene recognition more accurate and universal, we fused the recognition results of the above two models. In this work, $\vec{f} = \{f_1, f_2\}$ denotes factor vector, where f_1 and f_2 represent the weighted factors of environment model and image model, respectively. The weight factor value is different in different cases. Then, the linear weighting method is used to fuse the recognition results of the above two models at the decision level to obtain a new result. The final confidence $scene$ of the scene is expressed as follows:

$$scene = (\vec{gps}; \vec{image}) \cdot \vec{f} \quad (1)$$

Finally, we obtain the final scenario result $\max(scene)$ according to maximum principle.

B. Life Model Database

The life model actually represents a high-level scenario information database that is deployed on the intelligent cloud. The abundant computing and storage resources of the cloud allow the life model to store multidimensional scenario information. First, the intelligent cloud receives the data from the edge nodes. Raw data are obtained from the micro sensors, GPS, and other sensing devices. By using multiple deep learning algorithms, the system can extract high-level information

TABLE I
COGNITIVE PARADIGM EXAMPLES

Type	Example	Description
$paradigm_1$	$Average(A_{user,health}), A_{(user,age)} \in (age1, age2]$	$A_{user,age}$, $A_{user,hea}$ represent age and health status attributes in the user dimension
$paradigm_2$	$A_{envir,wea} = rain \rightarrow A_{user,moo}$ is mostly <i>Depress</i>	$A_{envir,wea}$ and $A_{user,moo}$ denote weather in the environment dimension and mood attribute in the user dimension
$paradigm_3$	$ShopEvent : \{(A_{envir,pos} : mall), (A_{beha,act} : look)\} \rightarrow \{(A_{beha,act} : dress)\} \rightarrow \{(A_{beha,act} : talk), (A_{user,escort} : friend)\}$	$A_{envir,pos}$, $A_{beha,act}$, $A_{user,escort}$ denote position, action, and partner attributes in the environment, behavior, and user dimensions

from the low-level data. For instance, a deep learning-based model can be used to realize individual tracking, and then activity information can be obtained from images or videos. Furthermore, the high-level information can be stored in the multidimensional datasheet in the life model database.

The life model contains multidimensional information, including information about the user, environment, time, society, and other fields. The type and the number of dimensions vary in different service scenarios. For a given service scenario, the datasheet is designed as follows. D_t represents the t dimension, and we also denote the attribute set of D_t . Then, $D_t = \{A_{t,1}, \dots, A_{t,n_t}\}$, where $A_{t,i}$ denotes the attribute value of the i_{th} feature in dimension t . Actually, the attribute value series in D_t summarizes the important state of person in dimension D_t . Take the three dimensions $user \times environment \times time$ as an example to explain the storage mode. These three dimensions have several attributes:

- User dimension: basic information about users, including name, age, occupation, address, and more.
- Environmental dimension: the physical environment of users, including the position, air quality, noise, traffic, temperature, weather, and more.
- Time dimension: the time information in the scene, including the date, season, week, time quantum, and more.

The idea of multidimensional data is used to store the user models. In the life model database, each attribute value is intuitively stored in a stereo structure block. The scene space $user \times environment \times time$ is presented in Fig. 2, where the 3D representation maps three datasheets in the database space depicting the 3D life model. The actual dimension design is based on different scenario categories. This is a full coverage database system. It can effectively organize users' life data in different time and space dimensions, providing an important data guarantee for information cognition.

C. Cognitive Paradigm in Database

The life model database mainly describes the dimension attributes. It also includes cognitive paradigms (see Table I) using mathematical statistics and reasoning rule methods. In the cognitive paradigms, the information is cognized and processed to get more value-intensive information.

A derivation paradigm obtains derived or inferred statistical information from basic data [29] based on statistical methods.

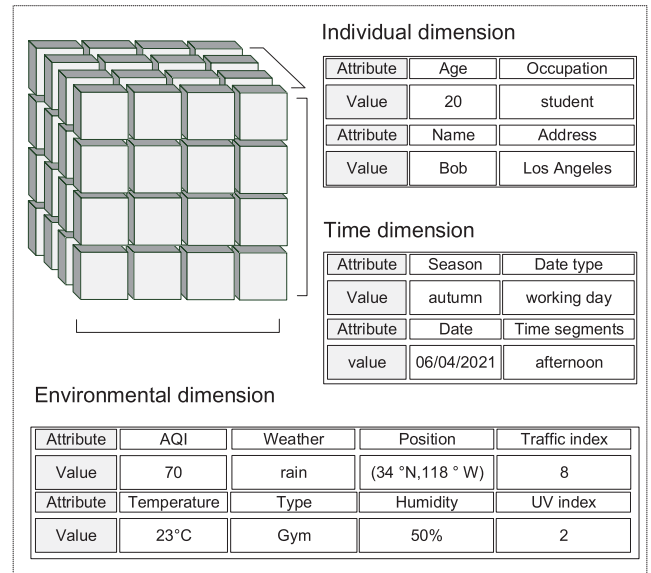


Fig. 2. Structure of 3D table space in living model database.

For example, the health condition of users in different age groups is analyzed in the user dimension. For an $[age_1 - age_2]$ age group, the computing paradigm for users' average health condition is $paradigm_1$. This kind of paradigm can be used for crowd simulation imitating universal life models of certain types of people.

Rule paradigms are actually rules that can infer useful results from the behavior of users. Such paradigms are reliable because they summarize a large quantity of facts. It is supposed that Bob is usually depressed when it rains. The life model database entry for of Bob can store a rule paradigm as $paradigm_2$. The rule paradigm shows the emotional reaction of the user toward weather. As can be seen from this example, rule paradigms vary for different users. Therefore, rule paradigms are suitable for detecting the emotional condition of a crowd.

An event paradigm is a record of the continuous actions of an event (i.e., a sequence of actions). Individuals in the environment will produce complex and diverse behavior patterns. For example, in a crowded place such as a train station, if an individual exhibits abnormal behaviors such as robbery or abnormal walking, they can be recorded by an event paradigm. The corresponding event paradigm is described

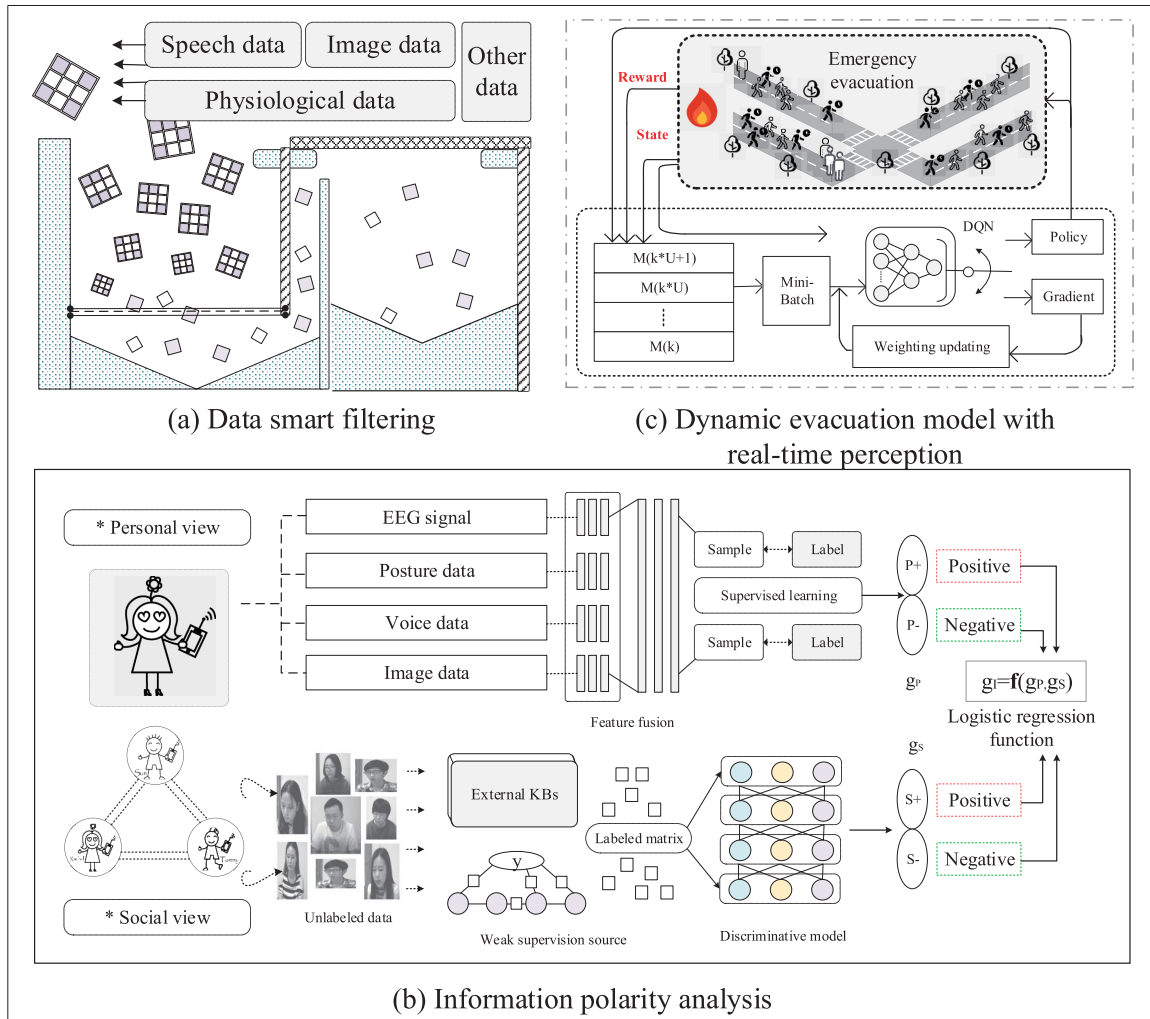


Fig. 3. Cognitive component structure containing cognitive information theory.

as *paradigm*₃. This kind of paradigm has a special incident feature. It is suitable for recording behavioral patterns along time and spatial dimensions.

IV. THE INFORMATION COGNITION COMPONENT

It is a challenge to scientifically transmit and apply the large quantity of data that flow in the proposed system. The goal of information cognition is to recognize the information after basic data preprocessing and further information extraction, so as to obtain higher value and more effective information. It is different from the analysis of data, which applies from the influence of information on people to conduct in-depth analysis. From the perspective of cognitive information, we discuss the design issues of each link of the system regarding the information flow. First many abnormal or lost data packets can occur in the data transmission process. However, these low-value density data occupy considerable transmission resources, and can even lead to the congestion of a communication system. Second, people's behaviors affect further their emotions, which in turn affect behaviors. Individual emotions and behaviors will affect crowd emotions and behaviors. To build a comprehensive and deep model the lives of individuals and

the crowd, we need to take into account the influence of behaviors and emotions on each other, especially the influence of individuals on the crowd. This requires "information polarity" analysis. Third, crowd information is complicated, and the real-time situation changes continuously over time. Therefore, the model is required to keep learning and updating according to new data so as to optimize its performance in real time, which is extremely important in regard to the application of crowd evacuation. On the basis of our the analysis of the above issues, we designed the intelligent data filtering, information polarity classification, and cognitive components of emotion recognition and interaction, which are deployed in the corresponding system links. In this way, we realized the intellectualization of a virtual agent with limited resources. Fig. 3 shows the design structure of these three cognitive components.

A. Smart Data Filtering

Smart filtering denotes the process of clearing deficient data and selecting valuable data from a raw dataset. Due to users' mobility, it is impossible to know the data of all the dimensions in every moment. It is assumed that $x_t^i = null$ exists in X_t . In our filtering process, a corresponding existence vector

$IsX_t = is_t^1, is_t^2, \dots, is_t^n | is_t^i \in [0, 1]$ is created according to each X_t . $W = w_1, w_2, \dots, w_n$ is a vector of information value weight. Therefore, the loss degree of dataset X_t can be defined as $loss_t = \sum is_t^i$. The value magnitude of X_t is $value_t = IsX_t \cdot W'$. In the process of filtering data, dataset X_t will be discarded as abnormal data when $loss_t$ or $value_t$ is lesser than the threshold of information loss degree or the threshold value magnitude, respectively. For different individuals, there are different values of $loss_t$ and $value_t$. In the process of long-term monitoring and personalized model training for each individual, personalized parameters will be gradually generated to guide the adaptive adjustment of these two values, which change with time and space.

B. Information Polarity Analysis

In this work, the role of data polarity classification is to judge human behavior as positive or negative. Positive helps produce positive emotion in people, making them feel happy and satisfied, and negative tends to make people depressed. Therefore, the analysis of information polarity can be transformed into the analysis of the emotional tendencies of humans. In the traditional classification of emotional tendencies, only individuals' emotional states have been considered. In this paper, information polarity is proposed based on two aspects: the individual's emotional state, and the positive or negative influence of personal behaviors on the development of human civilization. Personal behaviors influence the emotional performance of a society. Therefore, it is subjective to use only personal emotion states to represent the information polarity, so we analyze information polarity from two points of view: the personal view and the social view.

Personal-view information includes the EEG signal, behavior sequence, voice data, and facial images of users. The emotional tendency contained in the heterogeneous data, especially the EEG signal and behavior sequence, is not visual. The learning model can be used with supervision to judge the emotional tendency. In earlier studies, users self-evaluated their emotional tendencies. Then, the EEG data of each time slice were labeled. When the model converges, the classification model has been trained well. If the personal multimodal data are input into the classification model, the emotional confidence coefficient vector $g_P = \{p_+, p_-\}$ can be obtained.

The social view is a simple data type obtained mostly from the expressions of other people. Besides, it can be infeasible to require other people in society to emotionally evaluate the images. Therefore, the social view model uses a weak supervision method to classify the polarity of emotions from a social perspective. To solve for data without a label [30], first, the labeling function is used to label the image data automatically. The labeling function can be described as a model that generates labels by extracting relevant label information from image data found on the internet, in literature, and from open datasets using machine learning-based methods. Each unlabeled image has several labels attached by the labeling function, including a clean label and a noise label. The data and labels containing the noise are input into a judgment model, which infers the probability of unknown labels. Then,



Fig. 4. Experimental platform for crowd monitoring.

the expected loss value of the relevant probability judgment model is minimized, and data labels are denoised based on the emotion vector $g_S = \{s_+, s_-\}$.

After getting the information polarity vector (i.e., g_P, g_S) viewed by individuals and the society, the linear regression information polarity vector $g_I = \{i_+, i_-\}$ is obtained as follows:

$$i_+ = w_p * p_+ + w_s * s_+ \quad i_- = w_p * p_- + w_s * s_- \quad (2)$$

where w_p and w_s are the contributions of personal and social views to the integrated information polarity, and $w_p + w_s = 1$.

C. Dynamic Evacuation Model With Real-Time Perception

One important application of crowd monitoring and modeling is emergency evacuation. Traditional evacuation strategy is static and fixed, which means that the method of guidance (a broadcast or on-site staff) and evacuation routes are determined beforehand and can be used immediately when needed. However, fixed strategies cannot be optimized or updated according to real evacuation event experience. In a real evacuation, various kinds of unpredictable emergencies can occur. Therefore, a better evacuation model should be able to update itself in real time to adapt to real situations.

Considering this, we employed the Q-learning method of reinforcement learning theory to analyze the real-time situation and optimize the evacuation strategy [31].

The evacuation state at moment t is denoted as s_t . We can set the percentage of people not evacuated yet as different evacuation states. For example, we can set ten evacuation states. When 0%-10% of people are not yet evacuated, the current state is state1; when 90-100% of people are evacuated, it is state10. The evacuation strategy (guidance by broadcast or on-site staff, different evacuation routes, etc.) is denoted as a_t . $Q(s_t, a_t)$ and $R(s_t, a_t)$ represent the total estimated return and immediate award using the evacuation strategy a_t when the current evacuation state is s_t . For example, we can set the speed of evacuation (how many people were evacuated during the last time unit) as the immediate award $R(s_t, a_t)$. $Q(s_t, a_t)$ is updated by the temporal difference method, which is expressed as:

$$Q \leftarrow (1 - \alpha)Q + \alpha(R(s_t, a_t) + \gamma \max_{a'} Q(s_{t+1}, a')) \quad (3)$$

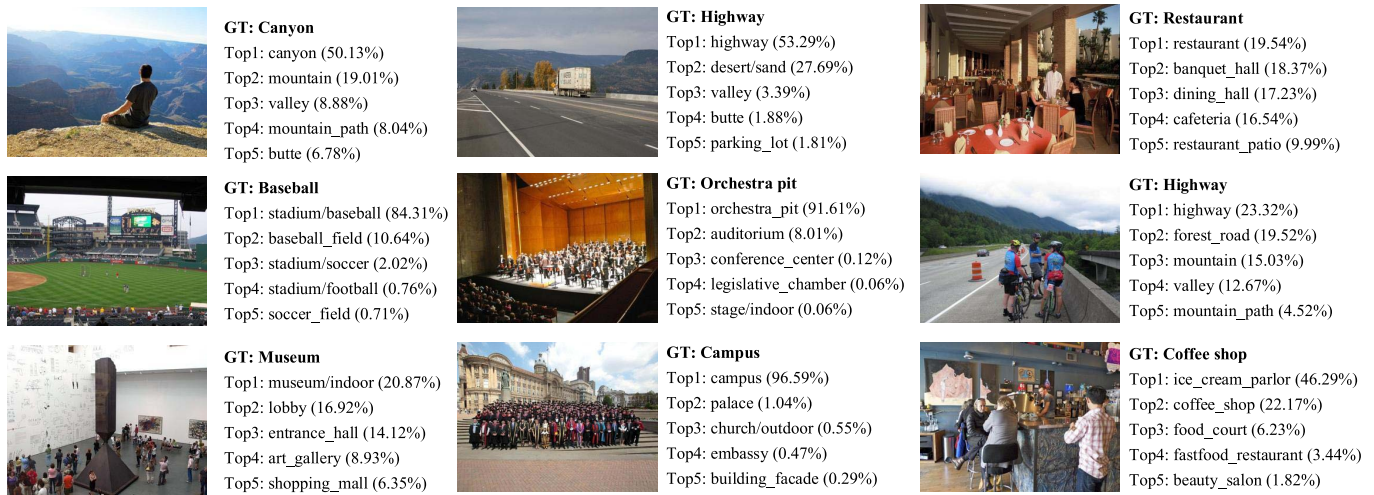


Fig. 5. Scene recognition results based on ResNet50.

where α denotes the learning rate, and γ denotes the reward disintegration coefficient. The initial value of $Q(s_t, a_t)$ is zero.

In this way, the dynamic evacuation model with real-time perception can be used to find whether the current evacuation strategy is suitable for the real time situation. The Q table will get a high or low return value according to the response. The model converges continuously as the dynamic evacuation mechanism improves.

V. EXPERIMENT

To verify the feasibility of the proposed ecosystem, we carried out experiments with three aspects: individual life modeling, scene perception, and crowd simulation.

As shown in Fig. 4, our life modeling platform includes four parts: a data collection layer centered on smart clothes [32], an edge service node, a big data platform deployed with the AI algorithm, and a visualization platform. The data collection layer is designed based on the concept of intelligent fabric smart clothes. The outfit used consists of an intelligent bolster, wearable smart clothes, and an EEG signal collection cap. The edge service node contained two local servers with Intel Core2 at 2.66 GHz, and 8GB RAM with Dual Inline Memory Module. A big data platform was used as a remote cloud and built based on Apache Hadoop distributed frame. The visualization platform is a crowd simulation platform that combines individual life modeling and scene perception. It was used to display crowd situations in specific scenes.

In particular, the data collection layer contained a voice sensor, temperature sensor, force sensor, and micro camera that were used to collect voice information, environmental temperature, embrace force, and facial expression, respectively. Flexible cables were used in the smart clothes to connect heart rate sensor, temperature sensor, blood oxygen detector, and some other micro sensors to collect users' physiological data. At the same time, the GPS sensor on the back of the smart clothes acquired position information on physical user location in real-time. The EEG signal collection caps collected a weak signal generated by the cerebral cortex including three electrode points. Two of them were used for calibration,

while the remaining one was used to gather single-channel EEG data. The local server was worked as a communication layer between the data collection layer and big data platform. At the same time, relevant data pre-processing algorithms were deployed to filter the data in the front end of the system. The big data platform had two management nodes and seven datum nodes. A few algorithm interfaces could load different algorithm models. Besides, a large amount of data were stored in the database, reflecting the multidimensional user. In the experiment, the crowd simulation platform was used to display the data collection results and the state of crowd simulation was used to realize the crowd monitoring.

Fig. 5 shows the results of scene recognition based on scene perception. We used ResNet50 to train the model on a public dataset then selected some parts of scene pictures for testing. This dataset covered about 365 scene categories. Our model accurately identified the place where crowds were located, whether in low-density or high-density scenes. Scene categories were relatively fine-grained, which greatly helped us realize the simulation system.

After scene recognition was completed, on the one hand, it was necessary to locate the people in the scene; on the other hand, modeling the two-dimensional image to the 3D world for crowd simulation needed to be implemented. Fig. 6 depicts the experimental results of crowd location and simulation modeling. In the first row are the original images with the scene recognition results and belief probabilities obtained by the ResNet50 network. In the second row are crowd positioning results obtained by the RAZ_Loc algorithm, and in the third row are the results of crowd simulation. The three original images came from a crowd dataset rather than a scene detection data set, so they did not have scene labels. Still, the scene recognition algorithm worked well in the three images. For each image, the top 1 predicted label (i.e., the label with the highest predicted probability) was similar to manual judgement results. As shown in the second row, the crowd positioning algorithm gave the 2D positions of each face accurately. Having the 2D positions, we conducted a simple 2D-3D conversion using the perspective transform function of

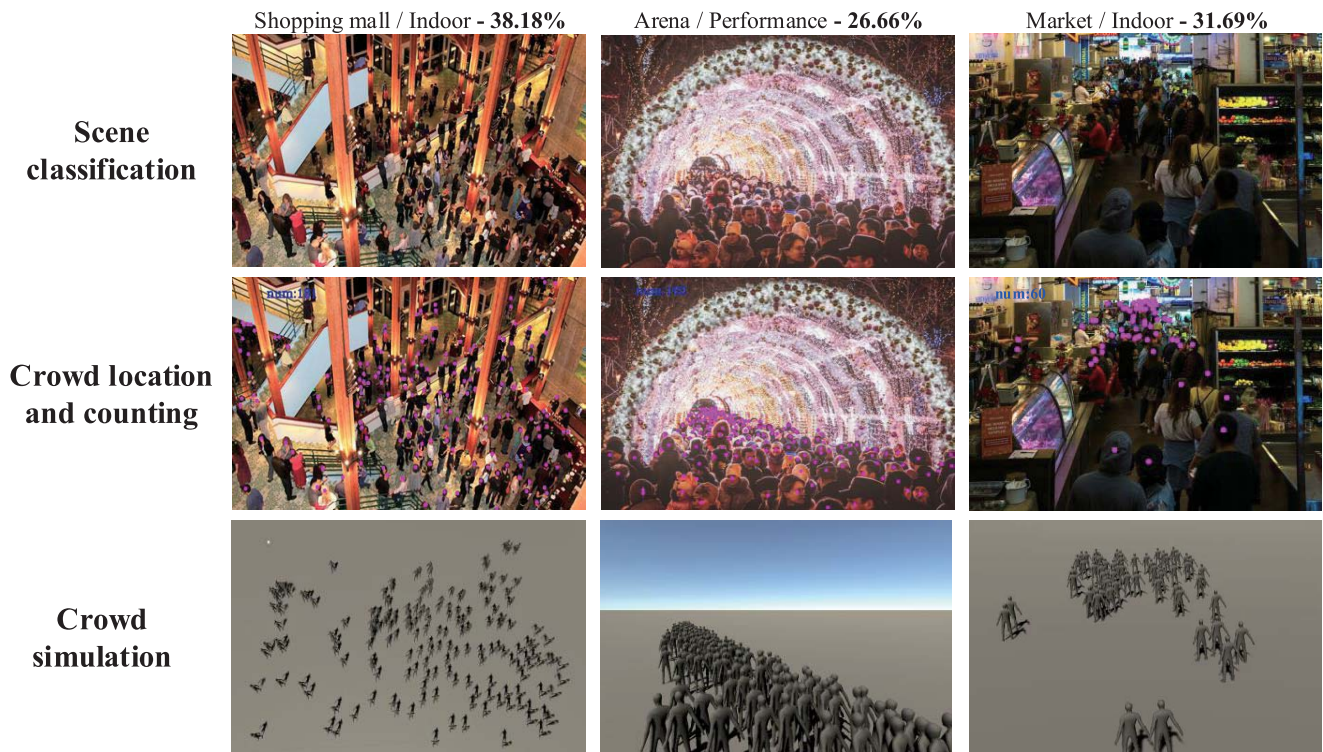


Fig. 6. Scene awareness results with belief probability and crowd location simulation.

OpenCV, (Python library). The parameters required by this function were selected manually since the images came from the Internet and we lacked camera parameters to conduct accurate conversion. In the crowd simulation outputs, everyone appears at the converted 3D positions, which looks similar to the original image.

Through the above experiments, we realized the cognition of the crowd's environment, the modeling and simulation of the crowd, and the life modeling of the crowd, all of which can be effectively used in our crowd monitoring system. Most of the ideas in this paper were verified. However, due to the limitations of the experimental conditions, it was difficult to carry out the experiment of human personalized all around modeling. In future research, more data related to human body should be collected to completely model of human life.

VI. CONCLUSION

In order to meet the challenge of low levels of intelligence in existing crowd monitoring systems, massive data redundancy and a lack of distinction among personalized information, we proposed a limo-powered crowd monitoring system to realize in-depth life modeling. A variety of sensors can collect individual data and model life situations, and realize cognition concerning individual members of crowds. On the other hand, it is possible to perceive crowded environments through intelligent algorithms, including scene recognition and crowd location. With the help of intelligent fabric, we designed an ecosystem to collect the all-around data humans and build a multidimensional human life model. Additionally, by considering the network communication of the whole system and cognitive information theory, we deployed cognitive

components in our system, and optimized the front-end data cognition process and back-end intelligent interaction process are optimized. According to our experimental results, an algorithm deployed in the edge cloud can use our system to realize intelligent crowd modeling and simulation, which is of great significance to the advancement of intelligent crowd monitoring.

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